

PRACTICAL 1 — File Copy using Linux System Calls

```
Hello, this is File 1.
This file is copied using Linux system calls.
```

```
GNU nano 7.2 filecopy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytes;

    source = open("File1.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open("File2.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        perror("Error creating destination file");
        close(source);
        return 1;
    }

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");
}

data value: 100
data address: 0x599276e1a2a0
saireddy@DESKTOP-HKQRL9G:~/COW_Practical$ mkdir FileIO_Practical
saireddy@DESKTOP-HKQRL9G:~/COW_Practical$ cd FileIO_Practical
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ nano File1.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ nano filecopy.c
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ nano filecopy.c
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ ^C
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ gcc filecopy.c -o filecopy
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ ./filecopy
file copied successfully.
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ ./filecopy
```

Check the files

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ ls
File1.txt File2.txt filecopy filecopy.c
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ |
```

Check copied content

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ cat File2.txt
Hello, this is File 1.
This file is copied using Linux system calls.
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ |
```

PRACTICAL 2 — I/O Redirection using dup2()

```
saireddy@DESKTOP-HKQRL9G: ~$ nano
GNU nano 7.2
Hello from standard input.
This is an I/O redirection experiment.
```

```
GNU nano 7.2
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input, output;
    char buffer[100];
    int n;

    input = open("input.txt", O_RDONLY);

    if (input < 0)
    {
        perror("Error opening input file");
        return 1;
    }

    output = open("output.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (output < 0)
    {
        perror("Error opening output file");
        return 1;
    }

    dup2(input, STDIN_FILENO);
    dup2(output, STDOUT_FILENO);

    close(input);
    close(output);

    while ((n = read(STDIN_FILENO, buffer, sizeof(buffer))) > 0)
    {
        write(STDOUT_FILENO, buffer, n);
    }

    return 0;
}
```

```
Hello, this is File 1.
This file is copied using Linux system calls.
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/FileIO_Practical$ cd ..
sairedy@DESKTOP-HKQRL9G:~/COW_Practical$ mkdir Redirection_Practical
sairedy@DESKTOP-HKQRL9G:~/COW_Practical$ cd Redirection_Practical
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ nano input.txt
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ nano redirect.c
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ gcc redirect.c -o redirect
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ ./redirect
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ cat output.txt
Hello from standard input.
This is an I/O redirection experiment.
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ |
```

Upgrade